Concept2Cure.RI

TECHNICAL WHITE PAPER

One governed chain from evidence to filing

An AI-native, 21 CFR Part 11-governed regulatory submission platform spanning medical device and drug pathways — from source intake through grounded drafting and assembly to validated agency transmission.

1.44M	705	359	13
LINES OF TYPESCRIPT	GOVERNED DATA TABLES	REGULATORY AI TOOLS	AGENCY GATEWAYS

Concept2Cure, Inc.
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Assessed against the `concept2cure-v2` engineering branch

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Quantitative figures in this paper were derived directly from the repository at the revision above. Forward-looking statements regarding roadmap, market conditions, and competitive position are estimates, not guarantees of outcome.

CONTENTS

What this document covers

A technical assessment of the platform as it exists in code today — its architecture, its intelligence layer, the compliance machinery that makes its output filing-ready, and an honest ledger of what remains before a first real submission.

A NOTE ON METHOD

Every quantitative claim in this paper is derived from the repository itself — file counts, schema declarations, route registrations, test files, and the platform's own readiness ledgers — rather than from marketing material. Where the platform is incomplete, this paper says so and names the owner of the gap. Regulatory professionals evaluate vendors by whether claims survive scrutiny; the same standard is applied here.

SECTION 01

Executive summary

Regulatory submission software is fragmented into four silos that no vendor spans. Concept2Cure.RI is built as the spanning layer — and the standards transition now under way is the moment when that becomes purchasable.

The market failure

Bringing a medical product to market requires assembling evidence, drafting an argument from it, packaging that argument to an agency's exact technical specification, and transmitting it through a validated channel — with every step attributable, auditable, and defensible years later. Today that work is split across four categories of vendor that do not talk to each other: **evidence and literature platforms**, **AI authoring startups**, **publishing and validation incumbents**, and **regulatory information management systems**. Sponsors buy three or four, and stitch the seams together with consultants and spreadsheets.

The AI authoring entrants stop at the document boundary — they produce prose, not a submission. The publishing incumbents validate and transmit, but their AI is a workflow bolt-on rather than a drafting engine. Nobody carries a claim from its source document through to a validated, signed, transmitted package.

What has been built

Concept2Cure.RI is a single governed system that spans that entire chain. Source documents enter a tenant-isolated evidence store; a retrieval-grounded assistant drafts against them with per-claim citations; the editor refuses to save content that lacks clause-level provenance; approval runs through an electronic-signature workflow with a hash-chained audit trail; and the resulting package is assembled to eCTD or eSTAR structure, validated, and handed to one of thirteen agency transport adapters.

5,053	3,846	1,759	557
TYPESCRIPT MODULES 1.44M LINES	API ROUTE HANDLERS ACROSS 389 MODULES	TEST FILES BEHIND 79 CI GATES	ROW-LEVEL SECURITY POLICIES VERIFIED

Why now

Three regulatory shifts converge in this window. FDA's **eSTAR v7.0** became mandatory for every 510(k) and De Novo on 3 August 2026. The **eCTD 4.0** transition runs 2026–2029 across PMDA, EMA, and FDA — the one moment in a decade when switching costs drop industry-wide, and only a handful of vendors have cleared the EMA pilot. And AI-derived content in submissions is being actively normalized by the agencies themselves, which moves the procurement question from *whether* AI may be used to *how the AI is governed* — a question this platform was architected around rather than retrofitted for.

The honest position

The platform is pre-general-availability. Its engineering substrate is deep and its compliance machinery is real and tested, but a first live filing depends on items that code cannot conjure: licensed agency templates and validator licences, gateway credentials, and a quality-system qualification signed by a competent person. Those are enumerated in Section 08 with their owners. The engineering thesis — that the spanning architecture is the hard part and it is built — is the investment case.

THE ONE-LINE THESIS

The first AI-native, Part 11-governed, end-to-end submission platform spanning both device and drug pathways — entering a market whose four incumbent silos each solve a quarter of the problem, at the precise moment a mandatory standards migration forces every sponsor to re-evaluate its stack.

SECTION 02

The problem and the market window

The submission market is not underserved — it is mis-served, by vendors each solving one quarter of a problem that only makes sense as a whole.

Four silos, no spanning player

A benchmark of the competitive field, conducted August 2026 across both the device and drug sides, found the same structural split on each. The categories are stable, the boundaries are sharp, and the seams between them are where sponsors lose time.

SILO	REPRESENTATIVE VENDORS	WHERE IT STOPS
Evidence & literature	Celegence CAPTIS, DistillerSR, CiteMed, Nested Knowledge, Basil Systems	Produces screened evidence, not a submission-shaped argument.
AI authoring	Weave Bio, Peer AI, Yseop, Narrativa, Complizen, Cruxi	Stops at the document boundary — no assembly, validation, or transport.
Publishing & validation	Lorenz docuBridge, Extedo, Certara GlobalSubmit, Ennov	Validates and transmits; AI is a bolt-on, not a drafting engine.
RIM & lifecycle	Veeva Vault RIM, Rimsys, RegDesk, Greenlight Guru	Tracks registrations and process; does not generate the content.

The closest attempts to bridge two silos are recent and partial: one RIM vendor added lightweight authoring in May 2026 with no evidence engine behind it; one AI authoring startup reached the drug workflow in April 2026 through a contract-research channel rather than a product. Neither spans the chain. Meanwhile at least one publishing incumbent has stated publicly that it is evaluating the future of its regulatory writing unit.

The timing: a mandatory migration reopens every stack

eSTAR v7.0 — mandatory since 3 Aug 2026

Every 510(k) and De Novo now flows through eSTAR. FDA technical-screening error rates fell from 20.5% to 5.4% under it. The *formatting* pain is shrinking; value is migrating to predicate strategy, testing rationale, and clinical evidence — the content problem.

eCTD 4.0 — the 2026–2029 window

PMDA mandatory April 2026, EMA around 2027, FDA around 2028–29. Every sponsor must migrate. Only about four vendors passed the EMA v4 pilot. This is the decade's one industry-wide drop in switching costs.

Gateway access is open

FDA's ESG NextGen retired the legacy WebTrader client in April 2025 in favour of a portal, REST API, and AS2. There is **no FDA certification programme for eCTD software** — validation criteria are published and a new entrant can offer direct submission on sponsor-held accounts after a short test phase.

AI is being normalized — and governed

FDA deployed its own generative system in 2025; FDA and EMA published joint AI guiding principles in January 2026; EU AI Act obligations phase in from August 2026. AI content is becoming acceptable, which turns *governance of the AI* into the procurement question.

What that combination means

A buyer who must migrate to a new submission standard anyway, who is now permitted to use AI but must prove how it is controlled, and who currently pays three vendors to cover one workflow, is a buyer whose stack is genuinely re-openable. That condition does not persist. It closes when the migration completes and the incumbents' AI bolt-ons mature.

Market facts in this section are drawn from a competitive benchmark synthesized from public agency notices and industry research in August 2026, retained with per-claim sources in the platform's engineering documentation. Vendor capability descriptions reflect publicly stated positioning at that date.

SECTION 03

Platform architecture

A single TypeScript monorepo organized in five layers, each of which the layer above can only reach through a governed interface.

Surface LAYER 5	React 18 + Vite single-page application. **Chat-first**: capability is invoked by asking the assistant, not by navigating to a screen per feature. The canonical workspace is an asymmetric three-zone grid — contextual tree, an intelligence stream, and a live-rendered artifact pane that never inherits chat state, so it can be isolated for Part 11 rendering guarantees. A registry of 28 governed components is the single source of truth; raw form elements are rejected by a CI gate.
Governance LAYER 4	The compliance spine, described in Section 05. Hash-chained audit, electronic signature with signing-authority resolution and version binding, document freeze and lock, clause-level lineage enforced in-transaction, and a default-deny authorization boundary across the API.
Intelligence LAYER 3	AnA, the tool-calling regulatory assistant (**359 registered tools** across 13 capability categories); the AI gateway that every model call must pass through; and the RIM, a deterministic non-LLM layer that accumulates regulatory patterns over time. Detailed in Section 04.
Domain services LAYER 2	**217 service domains** covering the regulatory subject matter: eCTD assembly and lifecycle operations, eSTAR generation, substantial-equivalence and predicate analysis, clinical evaluation and literature appraisal, biostatistics, CMC, pharmacovigilance, quality management, and the thirteen agency transport gateways. Exposed through **3,846 route handlers** in 389 modules.
Substrate LAYER 1	PostgreSQL with pgvector, addressed through Drizzle. **705 table declarations**; a fresh-database installer provisions **687 tables and 557 row-level security policies** from empty in one idempotent command, verified against a clean Postgres 16. Object storage for source documents, Redis for cache and queueing, and a real-time collaboration backend for the editor.

Design decisions that matter commercially

Fail-closed by default

The server refuses to boot without its secrets and names what is missing. The AI gateway refuses to serve with no provider configured. The readiness endpoint fails closed on an incomplete schema and names the missing tables. A regulated buyer's first question is what the system does when it is wrong.

One of everything

An enforced doctrine that there is never more than one implementation of a capability, backed by CI gates that reject duplicate table definitions, unreachable migrations, and parallel runtime paths. It is why a system of this size stays modifiable.

Refusal as a first-class result

When the platform lacks a licensed template, a dataset, or a credential, it says so explicitly rather than approximating. End-to-end journey tests assert honest refusals as passing outcomes. In this market, a confident wrong answer is a liability, not a feature.

WHY A MONOREPO OF THIS DEPTH IS THE ASSET

Regulatory subject matter does not compress. eCTD backbone construction, lifecycle operations, controlled vocabularies, region-specific validation profiles, substantial-equivalence logic, and thirteen distinct agency transport protocols are each irreducibly detailed, and each must interoperate with the governance layer to be usable. The volume of domain code is not incidental scale — it is the barrier a point-solution competitor would have to cross to span the same chain.

SECTION 04

The intelligence layer

Three components: an assistant that acts through governed tools, a gateway that no model call may bypass, and a deterministic layer that learns without a model at all.

AnA — the regulatory assistant

AnA is the product surface. Rather than a chatbot bolted beside a feature set, it is the invocation mechanism for the feature set: **359 tools** spanning submission and eCTD, evidence and intelligence, device and IVD, authoring, clinical, governance, quality management, CMC, nonclinical, biostatistics, and clinical pharmacology. Context is assembled from three memory layers before each turn — the working thread, project memory retrieved by vector similarity, and account context — and every call is recorded to an AI audit log.

> **Grounding is verified, not asserted**
>
> After each turn, a deterministic self-verification round — no additional model call — extracts the verifiable identifiers from the answer (trial registrations, PMIDs and DOIs, FDA submission numbers) along with any quoted source text, and checks each against the evidence the assistant actually gathered that turn. Claims absent from that corpus are surfaced as unsupported. This catches the failure mode that makes generative systems unusable in regulatory work: a fluent citation to a document that does not exist.

The AI gateway — a policy chokepoint

Every model call passes through one gateway, and a CI gate rejects code that attempts to bypass it. The gateway performs provider routing and failover, health tracking, cost accounting, and two evaluation passes before dispatch: a synchronous check covering token budget, role-aware prompt-injection scanning, blocked patterns and rate limits; and an asynchronous PII/PHI pass over outbound content that can allow, flag, redact, or block, producing a structured audit verdict that never carries the raw sensitive value.

> **Placement — the enterprise unlock**
>
> Every provider the gateway can reach is classified by three facts: the substrate it runs on (shared frontier API, frontier-in-your-cloud, or self-hosted), the regions it can be pinned to, and whether it carries a zero-data-retention guarantee. A tenant requiring EU residency can only be routed to a provider whose placement includes the EU; a tenant requiring zero retention only to a substrate contractually capable of it; an air-gapped tenant only to self-hosted inference. The resolved substrate and region are recorded in the audit log for every call. This is what makes a large pharmaceutical customer procurable, and it is architectural rather than configurational.

The RIM — accumulating judgment without a model

The Regulatory Intelligence Model sits beneath the assistant and is deliberately **not** an LLM. It observes regulatory work as it happens and aggregates signals — recurring deficiencies, reviewer triggers, data gaps, consistency issues, language tendencies — into durable, tenant-scoped patterns whose confidence rises with repeated observation but is bounded, so a single noisy loop can never assert certainty.

It is pure and deterministic — no network calls, no wall-clock dependency, identity derived from inputs — so the same call sequence always yields the same state, a prerequisite for anything influencing a governed document. On top of the pattern store sit a readiness-scoring engine, a recommendation engine, an evidence-confidence model, and change-impact propagation.

> WHY THE RIM IS THE COMPOUNDING ASSET
>
> Frontier model capability is available to every competitor on identical terms and improves for all of them simultaneously. A deterministic, tenant-scoped, versioned record of what reviewers actually objected to — accumulated across customers and submissions — is not purchasable. Its value grows with usage rather than with the next model release.

SECTION 05

Trust architecture

In regulated life sciences, output that cannot be defended under inspection has no value. The compliance machinery is not a feature tier — it is the reason the output is usable at all.

21 CFR Part 11 mechanisms

Hash-chained audit trail

Transactional SHA-256 hash chaining with anti-forgery locking, enforced by a database trigger on insert so the chain cannot be bypassed by an application path. A background integrity sweep runs by default in production. The chain is the primary evidence that a record was not altered after the fact.

Electronic signature

Signing requires credential re-entry. The signature path resolves signing authority and signer role, binds the signature to a specific document version, and persists the governed action. Frozen documents become immutable; locks prevent concurrent mutation of a record under review.

Clause-level lineage

The editor *cannot* save content without clause-level provenance — enforced in-transaction and structurally gated in CI, not by convention. A user can select any passage in a document and ask where it came from, and the system answers with the source, the model, and the audit identifier.

Tenant isolation

PostgreSQL row-level security with `FORCE ROW LEVEL SECURITY`, 557 policies applied and verified at install, an allowlist kept in sync by a CI gate, and a production posture that requires an explicit operator decision to activate rather than defaulting on silently.

Authentication and boundary

JWT with a pinned signing algorithm and CI-enforced zero-downtime key rotation, bcrypt password hashing at 12 rounds, mandatory attempt-limited email one-time-password challenge, and a global default-deny authorization boundary across the API surface so a newly added route is protected before anyone remembers to protect it.

The qualification pack

A generated installation- and operational-qualification pack maps eleven Part 11 controls to the specific tests that exercise them. All eleven execute and pass, and the pack's verdict **fails closed** on any control lacking evidence — it cannot report success by omission. What the pack cannot do is sign itself: a qualification is a protocol executed and signed by a competent person, and the pack is that person's input rather than a substitute for them. This is stated plainly in the platform's own ledger, and it is stated plainly here.

A DISCIPLINE WORTH NOTING

A CI gate scans the codebase for unqualified compliance claims. When an earlier build displayed "FDA 21 CFR Part 11 Compliant" on the public signup page, the platform's own gap register flagged it as a claim the system could not yet support, and it was softened to describe supporting Part 11 workflows. A vendor whose tooling polices its own marketing language is a vendor whose technical claims can be relied on — and in a market where a compliance overstatement can invalidate a customer's filing, that discipline is commercially material.

SECTION 06

Regulatory coverage

Three go-to-market journeys run over one shared chain. The chain is what competitors do not have; the journeys are what customers buy.

01	02	03	04	05	06	07
Intake	**Evidence**	**Draft**	**Review**	**Assemble**	**Validate**	**Transmit**
Canonical source identity, chunking, embedding, project-scoped isolation	Retrieval, literature appraisal, predicate and precedent analysis	Grounded authoring with per-claim citation and lineage gate	Threaded review, approval chain, freeze, electronic signature	eCTD or eSTAR structure, PDF/A rendering, checksum manifest	Structural and regional validation, dispatch-readiness gate	Governed transmission with pre-transmit checks and acknowledgement

The three journeys

JOURNEY	SCOPE	POSITION AGAINST 2026 TABLE STAKES
510(k) / De Novo GATED ON TEMPLATES	FDA device premarket notification. eSTAR generation, predicate identification, substantial-equivalence argumentation, filing readiness.	Substantial-equivalence engine, versioned template registry, and claim-to-source traceability are live and match or exceed the device-side field. Multi-user authoring with Part 11 audit is stronger than any device competitor assessed. Official template packs and field maps are a procurement gate the system fails closed on.
CER / PER UI WIRING IN FLIGHT	EU MDR and IVDR clinical evaluation. Annex XIV structure, MEDDEV 2.7/1 rev 4 conformance, literature screening, vigilance integration.	Rule packs, conformance validator, governed export renderers, literature screening with enforced exclusion rationale, and MAUDE/FAERS clients are real. Human-in-the-loop provenance is the decisive advantage — notified bodies reject pure-AI clinical evaluations. Remaining work is surfacing built services in the workbench.
IND / NDA VALIDATOR LICENCE PENDING	Drug submissions via eCTD. Backbone assembly, lifecycle operations, study tagging, US Module 1, PDF/A pipeline, gateway transmission.	eCTD 3.2.2 assembly is real with a qualification harness pinned to published FDA validation criteria, and eCTD 4.0 builders are carried. An FDA-criteria validation subset is built with an adapter seam ready for the industry-reference validator. Transport code is real; credentials are operational.

Agency transport

Thirteen submission-gateway adapters are implemented — FDA ESG, EMA CESP, PMDA, MHRA, Health Canada, TGA eBS, Swissmedic, NMPA, MFDS, ANVISA, CDSCO Sugam, HSA PRISM — behind a single governed transmit path with pre-transmit checks, bundle-integrity verification, and acknowledgement tracking. Because there is no FDA certification programme for eCTD software, direct gateway submission on sponsor-held accounts is reachable without an incumbent's permission.

THE STRUCTURAL POINT

A competitor entering from the authoring side must build assembly, validation, and transport before it can span the chain. One entering from the publishing side must build grounded drafting and the governance to make AI output defensible. Both must then do it twice — once for device, once for drug. That double crossing is the moat, and it is the reason a platform is worth more than the sum of its journeys.

SECTION 07

Engineering scale and defensibility

Scale alone is not an asset. Scale that stays modifiable, under mechanical quality enforcement, in a domain that cannot be compressed — that is an asset.

What exists, counted

DIMENSION	MEASURE	NOTE
TypeScript modules / lines	5,053 / 1.44M	Client, server, and shared contracts in one repository
Database table declarations	705	687 provisioned from empty by the fresh installer
Applied schema migrations	200	Reachability enforced by CI
API route handlers / modules	3,846 / 389	Behind a global default-deny boundary
Domain service directories	217	Regulatory subject matter, not framework code
Registered AnA tools	359	Across 13 capability categories
Test files	1,759	Unit, route, database, security, and journey suites
CI guard scripts / workflows	79 / 19	Governance, security, and schema-integrity gates
TypeScript error debt	2,598 → 0	Held at zero by a no-regression gate

Quality enforced by machine, not by intention

The 79 CI guard scripts are the unusual asset. They do not check style; they check architectural invariants that would otherwise decay silently — that no code bypasses the AI gateway, that the editor's save path cannot skip the lineage gate, that no duplicate table definitions exist, that migrations are reachable, that no mock data or development authentication path reaches production, that JWT verification stays algorithm-pinned, that interface compliance claims are qualified, and that the completion ledger cannot claim done without evidence. In a regulated product, this class of enforcement is the difference between a system that can be qualified and one that must be re-audited on every release.

Where the durable defensibility sits

1 · The spanning architecture

Intake through transmission, governed end to end, across device and drug. The volume of irreducible domain code is the barrier — and it is already paid for.

2 · Governed provenance

Clause-level lineage enforced in-transaction and a hash chain enforced by database trigger. Retrofitting this into an existing authoring product means rewriting its persistence layer.

3 · Outcome data

Pairs of submitted content and the agency's actual response — the strongest available moat and the only one no competitor can purchase. **Nothing captures it today**; it is the highest-leverage open item on the roadmap.

4 · Placement-aware inference

Residency, retention, and air-gap constraints resolved per tenant at routing time. The prerequisite for large-pharma procurement, and architectural rather than a deployment option.

THE HONEST ASYMMETRY

Items 1, 2 and 4 are built. Item 3 — the compounding data moat — is designed for but not yet capturing. That ordering is the correct one: the substrate that makes outcome data trustworthy had to exist before the data was worth collecting. But it should be read as a moat under construction, not a moat in place.

SECTION 08

Current state, roadmap, and risks

The platform maintains its own mechanically checked completion ledger, in which every open item names an owner. This section reproduces its substance.

Readiness by category

CATEGORY	STATE	WHAT IT MEANS
Core engineering substrate	BUILT	Schema, provisioning, auth, tenancy, gateway, audit chain, lineage gate — CI-enforced.
Intelligence layer	BUILT	359 tools, grounding verification, placement routing, RIM store and scoring engines.
510(k) journey	GATED	Engine complete; blocked on licensed eSTAR templates and field maps, which it fails closed on.
CER journey	WIRING	Services and validators real; screening and evidence surfaces being connected.
IND/NDA journey	GATED	Assembly and qualification harness real; validator licence and credentials are procurement.
Data lineage at the ends	OPEN	Strong from draft to approval; defects identified at the ingest and filing ends, with fixes scoped.
Outcome-data capture	OPEN	The compounding moat. Designed, not built.
Qualification signature	QUALITY	Evidence pack generated and passing; execution requires a competent person.

What stands between here and a first live filing

Engineering

- Complete the CER evidence workbench
- Repair the ingest-end extraction path
- Consolidate the audit substrate
- Build outcome-data capture
- Build template-change ingestion

Procurement

- eSTAR templates and field maps
- eCTD document type definitions
- Reference validator licence
- Agency gateway credentials
- Terminology dictionary licence

Quality & operations

- Execute and sign the IQ/OQ protocol
- Prove the login delivery path end to end
- Rehearse backup and restore
- Activate tenant isolation in production

Principal risks

Procurement dependency — several licensed assets are single-source and the platform correctly refuses to fake them, converting an engineering timeline into a commercial one. **Incumbent response** — the largest RIM vendors began shipping AI agents in 2026, and the window in which their AI is a bolt-on rather than a drafting engine is finite. **Breadth versus depth** — spanning device and drug is both the differentiator and the execution risk. **Qualification burden** — vendor validation evidence is a recurring operational cost, not a one-time task.

IN SUMMARY

What has been built is the hard part: a governed chain from source evidence to transmitted submission, with provenance and audit enforced structurally rather than by policy, across both regulatory worlds. What remains is substantially licences, credentials, signatures, and surface wiring. The market is fragmented, the standards migration has forced it open, and the architecture that spans it exists in code today.